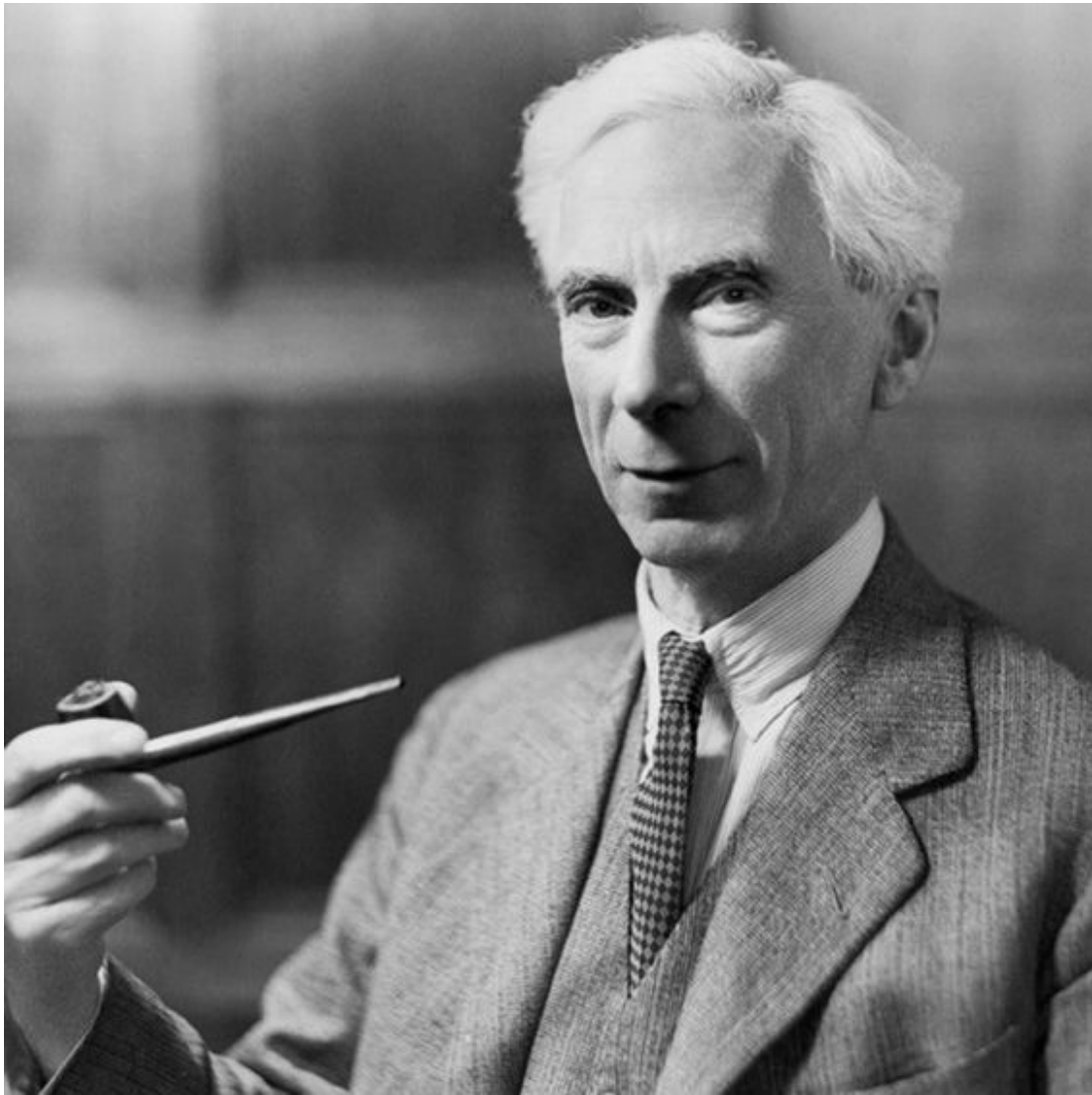


Principia Mathematica



Frege's work went virtually unnoticed outside a small circle of mathematicians, and was ignored by philosophers until it was discovered by Bertrand Russell. Russell had studied both philosophy and mathematics at Cambridge University and had independently concluded that arithmetic, and possibly all of mathematics, could be derived from logic, stating the case in *Principles of Mathematics* (1903). He and his colleague Alfred North Whitehead then attempted to prove this in the massive three-volume *Principia Mathematica*, completed ten years later. Following this, he turned his attention fully to philosophy, unsurprisingly concentrating on logic (and

probably influenced in this choice by his godfather, John Stuart Mill, who had written the definitive 19th-century book on the subject). Like Frege, Russell recognized the implications of breakthroughs in logic and mathematics for philosophy. They had shown that logic is an objective set of universal truths, and Russell believed that with science and philosophy both based on logic, we should be able to discover an objective knowledge of the world.